

SC FleetFinder — System Architecture Overview

A SaaS platform for coordinating multiplayer gaming group activities for the Star Citizen community.

Full-stack SaaS • Angular • Spring Boot • MySQL • Keycloak • Docker • Jenkins CI/CD • AWS EC2

Client Layer

Angular Frontend

- Typed REST API services
- Auth guards + interceptors (OIDC)
- Angular Material components
- Dev/prod environment configs
- Served via NGINX + TLS

Application Layer

Spring Boot Backend

- MVC structure
- DTO ↔ Entity mapping
- Validation + exception handling
- EhCache for reference data
- Role-based access + CORS
- Normalize user-entered timestamps to UTC

Data Layer

Database (MySQL + Flyway)

- Users, listings + 11 reference tables
- Flyway versioned migrations
- Testcontainers for DB testing

Identity Layer

Keycloak Identity

- Centralized login (OIDC)
- User roles + JWT-secured APIs

Deployment Layer

DevOps & Cloud

- Jenkins automated CI/CD
- Docker Compose (local & AWS EC2 prod)
- NGINX reverse proxy
- Health checks after deploy

Quality Assurance

Testing & Quality

- JUnit, Mockito, MockMVC
- Integration tests (Keycloak + MySQL)
- API contract tests